
SUBHAM DUTTA

Ph.D Candidate

Department of Earth Sciences,
Indian Institute of Technology Kanpur
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Summary

A postgraduate, currently enrolled as a Ph.D. student at the Indian Institute of Technology Kanpur, interested in low-temperature and environmental geochemical processes. I integrate field and analytical data from natural samples to decipher geochemical processes at the Earth's surface. I am committed to continuous learning and professional development.

Research Interests

Carbon cycling

Stable isotope Geochemistry

Soil Organic Geochemistry

Soil Carbon Dynamics

Education

Indian Institute of Technology Kanpur

Ph.D Candidate, Earth Sciences

Kanpur, India

Jan 2022 – Present

- CGPA: 8.5/10

Indian Institute of Technology Kanpur

M.Tech., Geological Technology

Kanpur, India

July 2019 – June 2021

- CGPA: 8.25/10

University of Calcutta

M.Sc., Applied Geology

Kolkata, India

Sep 2017 – June 2019

- First Class, 75.5%

Sidho Kanho Birsha University

B.Sc., Geology (Hons.)

Puruliya, India

Aug 2014 – Aug 2017

- First Class, 72.25%
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Fellowships & Awards

Institute Fellow, IIT Kanpur

2022-Present

GATE, MHRD

2019-2021

Publications

- Jangir, Y., & **Dutta, S.** Planetary analog sites in the Indian subcontinent and the Indian Ocean: Underexplored environments suited for astrobiological and space research. *Frontiers in Astronomy and Space Sciences*, 12, 1712191. doi: 10.3389/fspas.2025.1712191
- Nizam, S., Acharya, T., **Dutta, S.**, & Sen, I. S. (2023). Occurrence, sources, and spatial distribution of fluoride in the Ganga alluvial aquifer, India. *Environmental Geochemistry and Health*, 45(5), 1975-1989. doi: 10.1007/s10653-022-01319-4

- Nizam, S., **Dutta, S.**, & Sen, I. S. (2022). Geogenic controls on the high levels of uranium in alluvial aquifers of the Ganga Basin. *Applied Geochemistry*, 143, 105374. doi: 10.1016/j.apgeochem.2022.105374
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Teaching Assistantship

Geology of Fuels

2024 – 25 / II

ES454A

- This is an undergraduate introductory course on Fuel Geology. I am responsible for the student evaluation (assignments, mid-semester, and end-semester) process.

Physical Hydrology

2024 – 25 / I

ES202

2025 – 26 / I

- This is an undergraduate introductory course on Physical Hydrology. I was responsible for the student evaluation (assignments, mid-semester, and end-semester) process.

Geochemistry

2023 – 24 / II

ES413A

2022 – 23 / II

- This is an undergraduate introductory course on Geochemistry. I was responsible for the student evaluation (assignments, mid-semester, and end-semester) process.

Isotope Geochemistry and Applications

2023 – 24 / I

ES649A

- This is a postgraduate advanced course on Isotope Geochemistry. I was responsible for the student evaluation (assignment, laboratory work, mid-semester, and end-semester) process.

Experimental Rock Mechanics and Rock Physics

2020 – 21 / II

ES657A

- This is a laboratory-intensive postgraduate course. I oversaw the laboratory classes and taught the sample preparation and experiments.

Environmental Geochemistry

2020 – 21 / I

ES644A

- This is a postgraduate advanced course on Environmental Geochemistry. I was responsible for the student evaluation (assignment, laboratory work, mid-semester, and end-semester) process.
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Experience with Instrumentation

- Ion Chromatograph (independent operator)
 - QQQ-ICPMS (independent operator)
 - UV-VIS Spectrophotometer (proficient)
 - Picarro G2201-i CO₂ and CH₄ isotope analyzer (proficient)
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Software Skills

- QGIS, ArcGIS (experienced)
- Origin-Pro (experienced)

- MATLAB (proficient)
 - Python and R (proficient)
 - Inkscape, Adobe Illustrator (proficient)
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Fieldwork

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| • Sundarbans, West Bengal, India | March 2025 |
| • Ganga Upstream Catchment, Uttarakhand, India | June 2024 |
| • Ladakh, India | May 2023 |
| • Gorson Bugyal, Auli, Uttarakhand, India | April 2023 |
| • Kanpur, India | September 2020 |
| • Eastern Ghats, India | January 2019 |
| • Similipal and Chandipur, Odisha, India | January 2018 |
| • Ghatshila, Jharkhand, India | December 2016 |
| • Bankura, West Bengal, India | December 2015 |
| • Puruliya, West Bengal, India | December 2014 |
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Industrial Training

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| • Dalli Rajhara Iron Ore Mines, Bhilai Steel Plant, SAIL, Chattisgarh, India | July 2018 |
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References

Dr. Indra Sekhar Sen

Associate Professor

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Dr. Rajib Kar

Professor

Department of Geology, University of Calcutta

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